

**SUPPLEMENT:
THE DEVELOPMENT OF RISK BEHAVIORS AND THEIR CULTURAL
TRANSMISSION**

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In this document, we present supplementary information for the study “The development of risk behaviors and their cultural transmission”.

- (5) If k agents in P_i become newly ruined, update belief distribution pessimistically:

$$b_{i,t} = b_{(i,t,k)} = (1 + \delta_i)^k \cdot b_{(i,t,0)}$$

- (6) Update beliefs $\Gamma_{i,t+1}$ based on the gambling outcome:

$$\Gamma_{i,t+1}|U_t \sim \text{Beta}\left(a_{i,t} + U_t, b_{i,t} + (1 - U_t)\right)$$

Fitness calculation:

- (1) At the end of each generation, for each agent i , calculate fitness:

$$\mathcal{F}_i = \exp\left(\frac{g_{i,T} - g_{i,0}}{T}\right)$$

- (2) Normalize fitness weights:

$$W_i = \frac{\mathcal{F}_i}{\sum_{j=1}^N \mathcal{F}_j}$$

- (3) If $\mathcal{G} = \mathcal{G}_{\max}$, stop.

Cultural selection:

- (1) Spawn N next-generation agents keeping the distribution of class markers constant by sampling them from previous generation without replacement. Each new agent inherits strategies based on parental fitness weights W_i , sampling only from individuals within their class.
- (2) For each strategy component $L_{i,j}$ in vector $\mathbf{L}_i$:
 - (a) Inherit with probability from parental distribution at random with weights W .
 - (b) Continuous strategies mutate with probability m_c , undergoing Gaussian perturbation with noise σ_c , clamped within $[0, 1]$; discrete strategies mutate with probability m_d , and when they do they are randomly selected from feasible sets ($\{\text{UB}, \text{PB}\}$ for learning bias and $\{0, 1\}$ for parochialism).
- (3) Set $\mathcal{G} = \mathcal{G} + 1$ and go back to **Initialization step 2.(c)**.

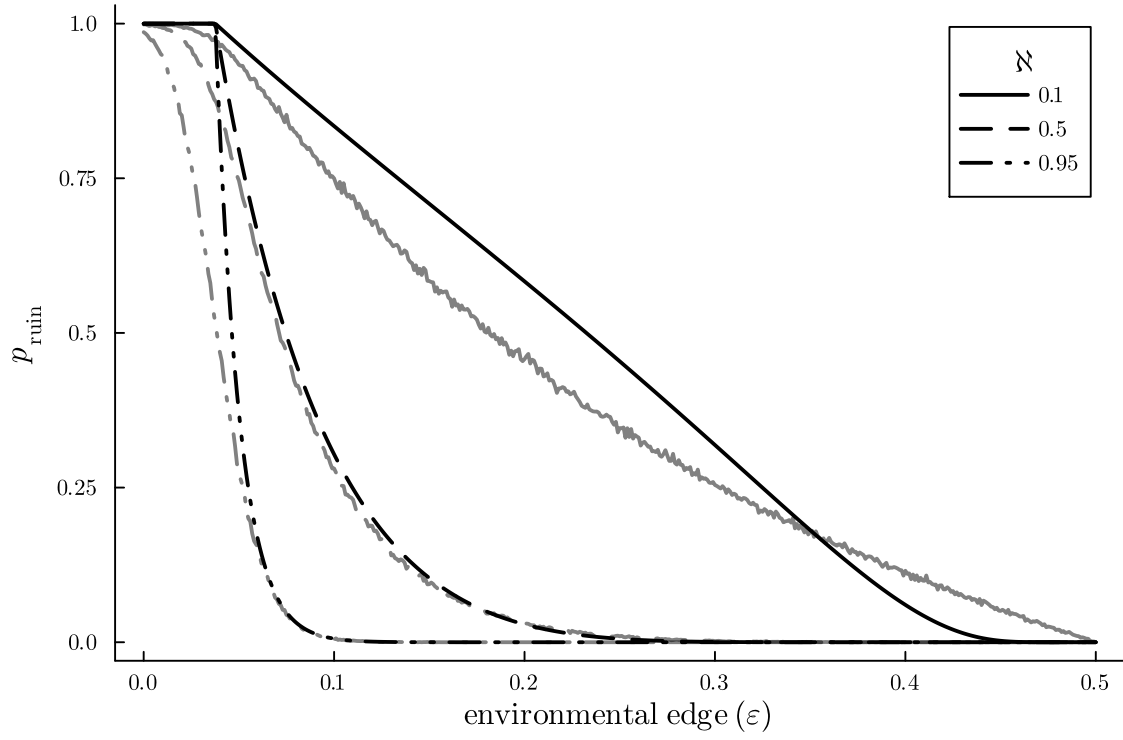


FIGURE S1. **The probability of ruin.** Plotted lines showing the simulated probability of ruin (grey) versus the one predicted by the approximation in the main text (black). $10e^4$ agents were simulated gambling for 1000 time periods using stakes of $s = 0.15$, and the number of ruined agents at the end of these periods is used to calculate the proportion of ruined agents, which is an estimator of the probability of ruin.

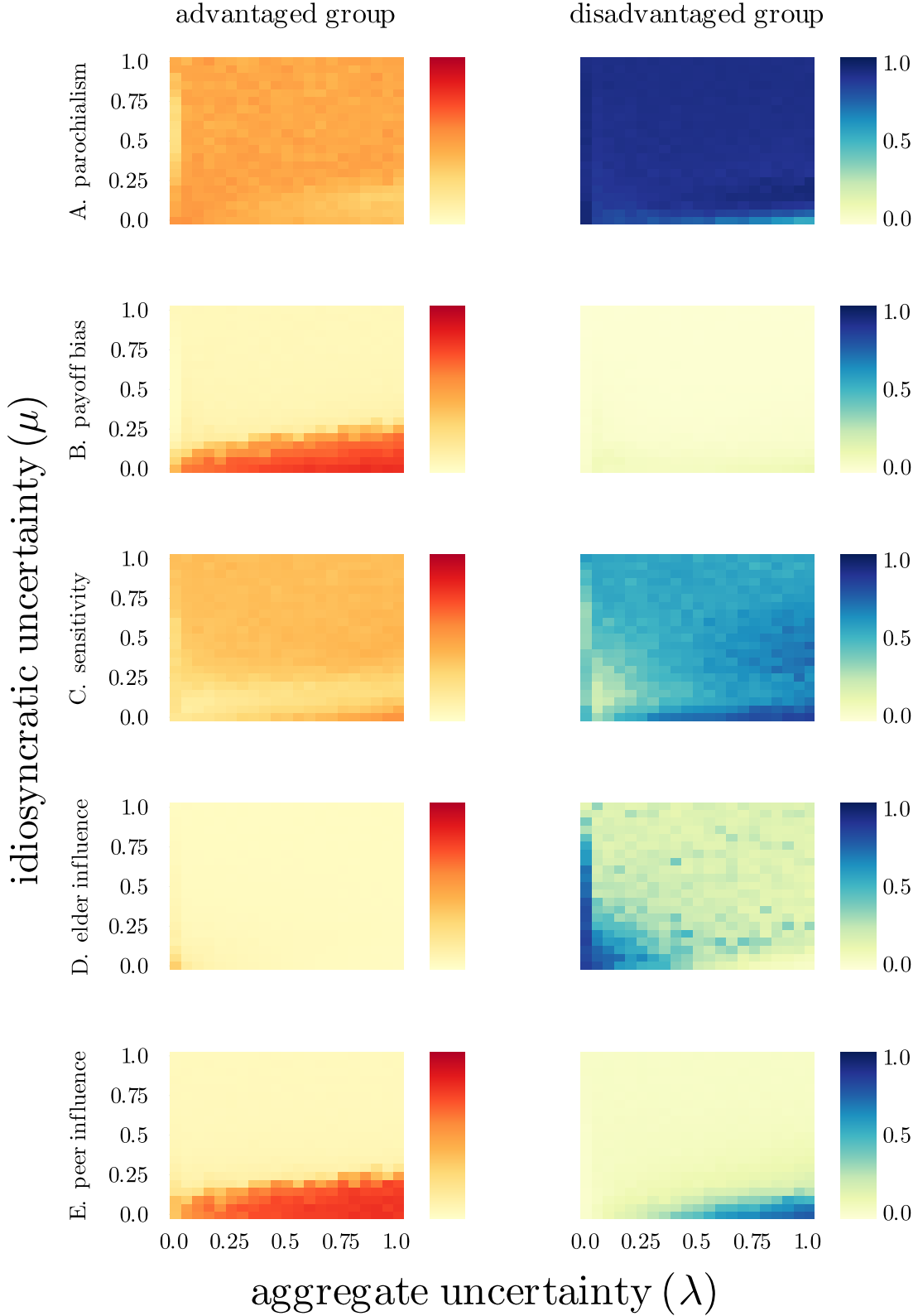


FIGURE S2. **Effects of class inequality and parochialism on learning strategies and risk behaviors.** Two classes, c_0 (red) and c_1 (blue), face distinct wealth buffers $\aleph_{c_0} = 0.95$ and $\aleph_{c_1} = 0.45$. Populations are initialized with $f = 0.5$. Each panel shows median evolved social learning strategy values for combinations of aggregate (λ) and idiosyncratic (μ) uncertainty values, which are varied from 0 to 1 in steps of 0.05. Initial environmental edges are sampled uniformly at random from the interval $(0, 0.5]$.

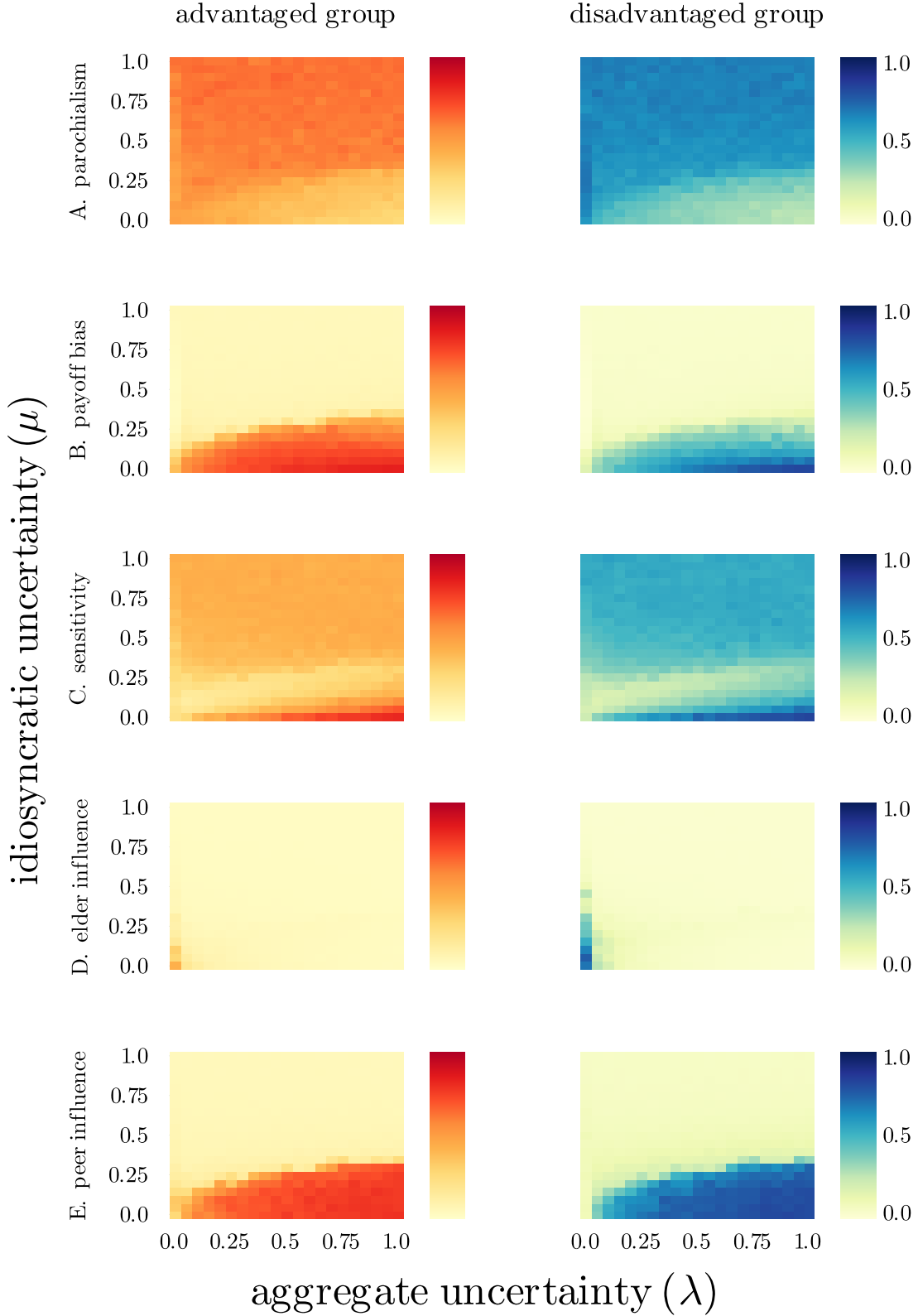


FIGURE S3. **Effects of class inequality and parochialism on learning strategies and risk behaviors.** Two classes, c_0 (red) and c_1 (blue), face distinct wealth buffers $\aleph_{c_0} = 0.95$ and $\aleph_{c_1} = 0.85$. Populations are initialized with $f = 0.5$. Each panel shows median evolved social learning strategy values for combinations of aggregate (λ) and idiosyncratic (μ) uncertainty values, which are varied from 0 to 1 in steps of 0.05. Initial environmental edges are sampled uniformly at random from the interval $(0, 0.5]$.

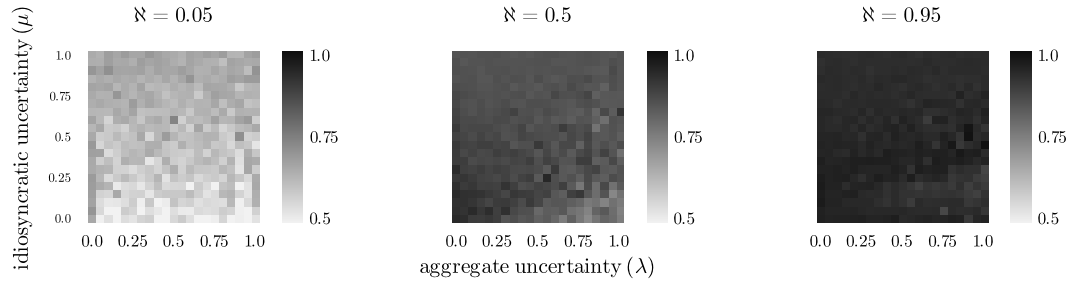


FIGURE S4. Mean stake change as a function of uncertainty. Mean lifetime proportional change in stakes relative to juvenile stake under aggregate (λ) and idiosyncratic (μ) uncertainty for low, intermediate and high values of N . Median values for proportional stake change are plotted as both μ and λ are varied from 0 to 1 in steps of 0.05. Initial environmental edges are sampled uniformly at random from the interval $(0.0, 0.5]$.

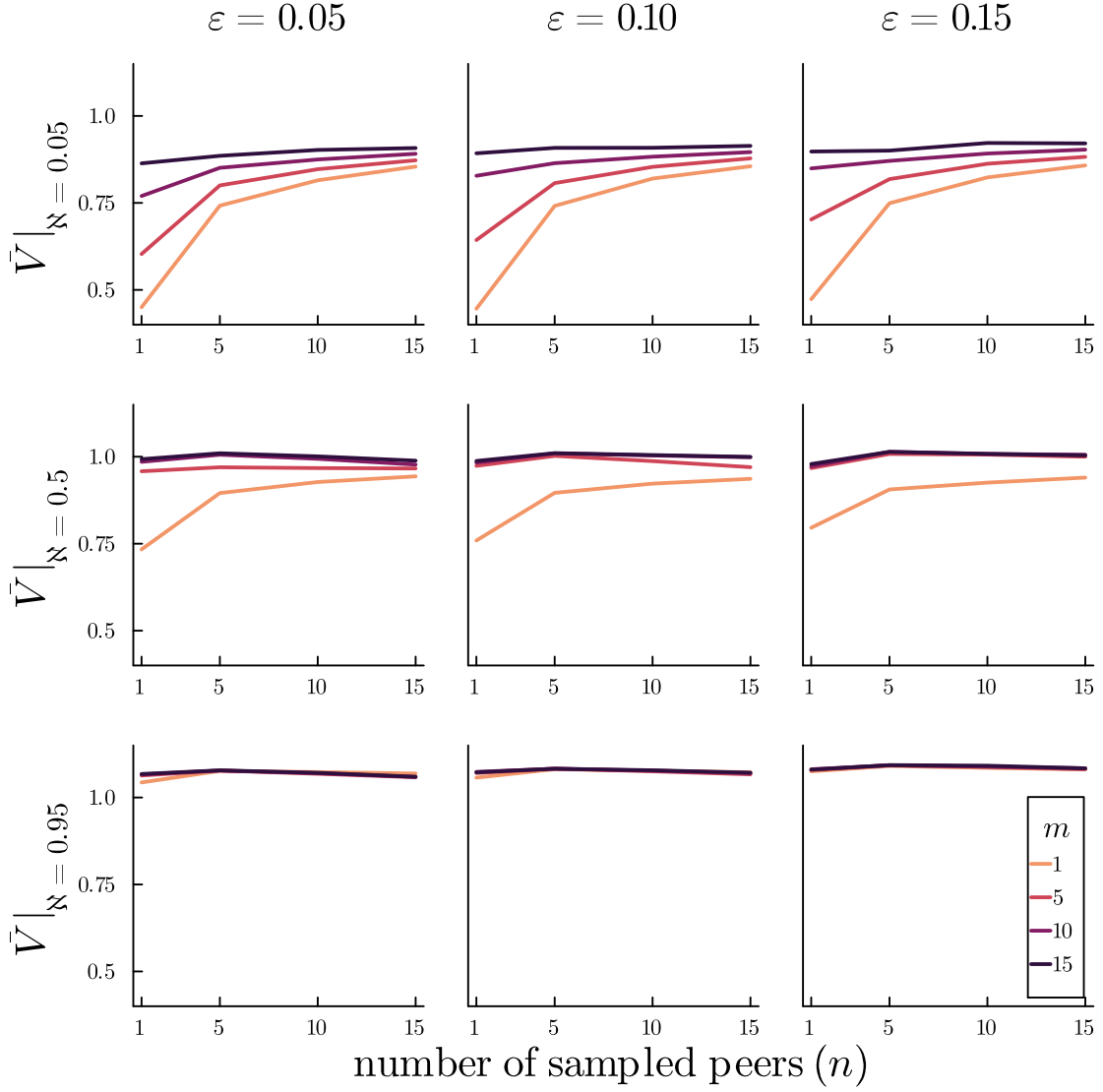


FIGURE S5. **Diminishing effects of number of sampled individuals on payoff.** Here we simulate average growth rates at equilibrium while varying the number of peers and elders to show that our results in the main text are robust to choices of n and m . Growth rates shown are averages across populations of $N = 1000$ agents taken at $T = 2500$. Increasing the number of sampled elders m leads to monotonic increases in average growth rate, while the relationship between average growth rates and number of sampled peers n is non-monotonic for intermediate wealth buffers ($N = 0.5$). However, the effects of n are indirectly controlled for by the evolution of the weight of peer influence parameter β , such that the differences in growth rate across different values of n remain very small.